LSPIM: 3GEE + FW versus Logistic + 3FW

Setup

Two engines of the LSPIM test are compared on the same generated datasets (generation run `e64245fe0a9d0284`; the analysis runs are `c1feac084af6cb59` and `22612da3a9139167`, produced by `scripts/run_lspim_engine_comparison.R`, where the engines are called `pgee_fw` and `geessbin`).

- **3GEE + FW** (short label **3GEE**, the original engine): three PGEE fits with `geessbin`, one for each clustering (left subject C1, right subject C2, pair C3), each with Ford-Westgate (FW) standard errors.
- **Logistic + 3FW** (short label **Logistic**): one penalised logistic fit and three FW sandwiches.

Under an independence working correlation the GEE estimating equations are the logistic score equations. The PGEE point estimate is therefore a Firth-type penalised logistic regression on the 0 / 0.5 / 1 pseudo-scores, with the penalty weighted by the Pearson scale ϕ (re-estimated at each iteration). It is not a plain `glm()`: the penalty term is what makes it PGEE. The clustering does not enter the equation for β, so the three 3GEE fits give the same β and only the sandwiches (C1, C2, C3) differ. Logistic + 3FW fits β once and computes the three FW sandwiches. The test is the same for both: Holm adjustment on the treatment-by-visit contrasts.

The scenarios are $n = 10, 20, 50$ clusters, three observations minimum per cluster, with a linear time trend ($\beta_3 = 0.22$ and $\beta_3 = 0$) and a log time trend (large crossing, $\beta_1 = -2.42$, $\beta_3 = 1.95$; and identical arms as null). That gives 12 scenarios with 5000 replicates each, tested at $\alpha = 0.05$.

"Has a decision" means the replicate gave a test result (no error and a non-missing rejection).

Decision agreement

Table 1: Decisions on the replicates where both methods give a decision (pairs).

Effect	n	Pairs	Both reject	Only 3GEE	Only Logistic	Neither	Agreement
Linear 0.22	10	3915	512	0	0	3403	1.00
Linear 0.22	20	4993	606	0	0	4387	1.00
Linear 0.22	50	5000	527	0	0	4473	1.00
Linear null	10	4000	474	0	0	3526	1.00
Linear null	20	4998	381	0	0	4617	1.00
Linear null	50	5000	183	0	0	4817	1.00
Log large	10	3933	854	0	0	3079	1.00
Log large	20	4999	1230	0	0	3769	1.00
Log large	50	5000	2737	0	0	2263	1.00
Log null	10	3971	462	0	0	3509	1.00
Log null	20	4992	407	0	0	4585	1.00
Log null	50	5000	215	0	0	4785	1.00

- Over 55,801 pairs in 12 scenarios, the number of replicates where only one method rejects is 0; agreement is 1 in every scenario.

Rejection rates

The rate columns are: **Own**, each method over its own replicates with a decision (as in the main simulation); **Common**, each method over the replicates where both gave a decision, computed separately for each method from the per-replicate rows; and the rate of 3GEE on the replicates where only Logistic failed (number of replicates in brackets).

Table 2: Type I error rate (nominal 0.05).

Effect	n	Own 3GEE	Own Logistic	Common 3GEE	Common Logistic	3GEE, only Logistic failed (n)
Linear null	10	0.153	0.119	0.118	0.118	0.291 (993)
Linear null	20	0.0766	0.0762	0.0762	0.0762	1.00 (2)
Linear null	50	0.0366	0.0366	0.0366	0.0366	-
Log null	10	0.146	0.116	0.116	0.116	0.262 (1027)
Log null	20	0.0824	0.0815	0.0815	0.0815	0.625 (8)
Log null	50	0.0430	0.0430	0.0430	0.0430	-

Table 3: Power.

Effect	n	Own 3GEE	Own Logistic	Common 3GEE	Common Logistic	3GEE, only Logistic failed (n)
Linear 0.22	10	0.169	0.132	0.131	0.131	0.312 (1055)
Linear 0.22	20	0.122	0.121	0.121	0.121	0.714 (7)
Linear 0.22	50	0.105	0.105	0.105	0.105	-
Log large	10	0.248	0.217	0.217	0.217	0.368 (1025)
Log large	20	0.246	0.246	0.246	0.246	-
Log large	50	0.547	0.547	0.547	0.547	-

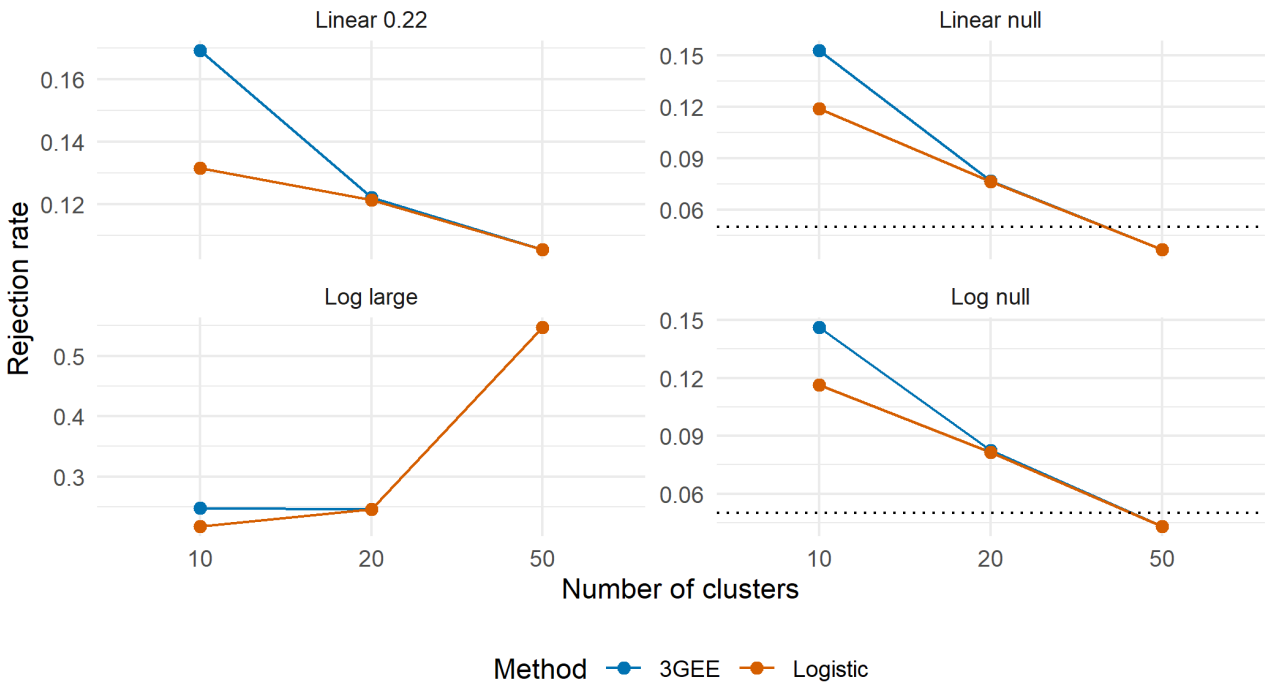


Figure 1: Rejection rate over each method's own replicates (dotted line: nominal 0.05 in the null panels).

- On the common replicates the rates of the two methods are identical in every scenario.

- At $n = 20$ and 50 the own-replicate rates are the same or nearly so, since nearly every replicate has a decision from both methods.
- At $n = 10$ the type I error of 3GEE is higher: 0.153 against 0.119 (linear null) and 0.146 against 0.116 (log null). On the common replicates the two rates are equal; the gap comes from the replicates that only Logistic + 3FW refuses (last column).

Failures and warnings

Table 4: Failures per scenario (5000 replicates each). Err.: no decision. Not conv.: a decision flagged as not converged. Only X: replicates where just X gave no decision.

Effect	n	Err. 3GEE	Err. Logistic	Not conv. 3GEE	Not conv. Logistic	Only 3GEE	Only Logistic
Linear 0.22	10	30	1062	0	23	23	1055
Linear 0.22	20	0	7	0	0	0	7
Linear 0.22	50	0	0	0	0	0	0
Linear null	10	7	994	0	6	6	993
Linear null	20	0	2	0	0	0	2
Linear null	50	0	0	0	0	0	0
Log large	10	42	1037	0	30	30	1025
Log large	20	1	0	0	1	1	0
Log large	50	0	0	0	0	0	0
Log null	10	2	1027	0	2	2	1027
Log null	20	0	8	0	0	0	8
Log null	50	0	0	0	0	0	0

The replicates where only one method fails fall into the following types.

- **Logistic + 3FW stops** on 4117 replicates where 3GEE returns a decision: 1954 with a leverage-1 cluster in clustering C1, 1995 with a leverage-1 cluster in C2 (the Ford-Westgate correction is undefined), and 168 with an all-zero design column (late `trt_visit` columns with no comparable pairs).
- **3GEE stops** ("infinite or missing values in 'x' ") on 62 replicates, 61 of them at $n = 10$. On these Logistic returns a decision, flagged not converged on 62 of them, with the warning "fitted probabilities numerically 0 or 1" on 62.

Table 5: Replicates where only Logistic failed, by failure type, and the rejection rate of 3GEE on them in the null scenarios (nominal 0.05).

Logistic failure type	Replicates	Of which in null scenarios	3GEE rejection rate, null scenarios
Leverage 1, C1	1954	953	0.264
Leverage 1, C2	1995	1004	0.282
All-zero design column	168	73	0.411

On the replicates that Logistic refuses with a leverage-1 error, 3GEE rejects the null in 0.251 to 0.298 of the cases per null scenario at $n = 10$, far above $\alpha = 0.05$.

Table 6: Share of replicates with a decision that carry a warning: combined V repaired to the nearest PSD matrix, and fitted probabilities numerically 0 or 1.

Effect	n	Nearest PSD 3GEE	Nearest PSD Logistic	Probs. 0 or 1 3GEE	Probs. 0 or 1 Logistic
Linear 0.22	10	1.00	1.00	0	0.00584
Linear 0.22	20	1.00	1.00	0	0
Linear 0.22	50	0.0106	0.0106	0	0
Linear null	10	1.00	1.00	0	0.00150
Linear null	20	1.00	1.00	0	0
Linear null	50	0.00980	0.00980	0	0
Log large	10	1.00	1.00	0	0.00757
Log large	20	0.999	0.999	0	0.000200
Log large	50	0.00600	0.00600	0	0
Log null	10	1.00	1.00	0	0.000503
Log null	20	0.999	0.999	0	0
Log null	50	0.00960	0.00960	0	0

Computation time

Table 7: Median computation time per replicate (seconds), median over the four scenarios of each n.

n	3GEE	Logistic
10	0.798	0.00900
20	2.69	0.0140
50	40.0	0.0630

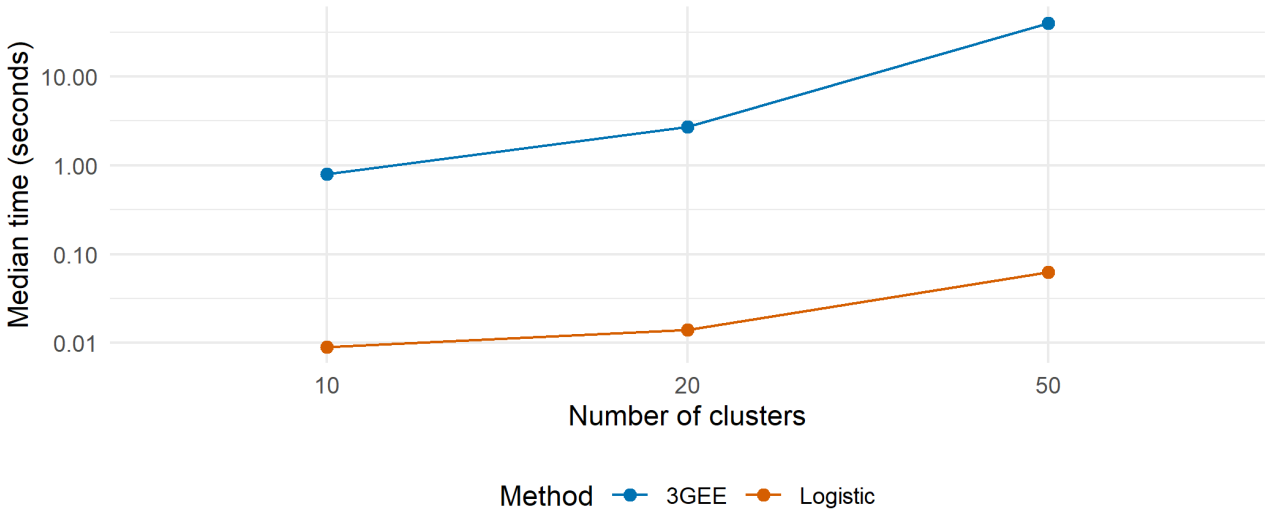


Figure 2: Median computation time per replicate (seconds, log scale), median over the scenarios of each n.

Conclusion

- Decisions are identical wherever both methods give one; the rates on the common replicates are equal; the own-replicate rates are identical at $n = 50$ and nearly identical at $n = 20$.
- The $n = 10$ difference comes entirely from replicates where Logistic + 3FW stops (a leverage-1 cluster or an all-zero design column) and 3GEE still returns a decision; on those 3GEE rejects far more often than α under the null.

- One earlier check (2026-10-08 scratch verification, not part of this run): in such a case 3GEE reported "converged" with a C2 variance of 3.4e-17 for the affected column. This is a single example.
- Logistic + 3FW is several hundred times faster (about 635 times at $n = 50$).